



User Acceptance in Human–Robot Interaction: Exploring the Role of Anthropomorphic Mechanisms in Manufacturing Environments—A Systematic Literature Review

Ana Pinto^{1,2,3} · Andrey Solovov³ · Ana Correia Simões⁴ · Paulo Menezes³

Received: 19 July 2024 / Accepted: 20 March 2025 / Published online: 14 April 2025
© The Author(s) 2025

Abstract

In pursuing Industry 5.0's vision, which emphasises human well-being and the seamless integration of robots into manufacturing processes, understanding the role of anthropomorphic design is crucial. Anthropomorphic design, where robots exhibit human-like, animal-like, or even entirely novel traits (e.g. a display scrolling text), aims to improve human–robot interaction (HRI) and enhance human acceptance within manufacturing contexts. Understanding the optimal degree of human-readable characteristics in robots is essential for further advancements in this domain. This systematic literature review aims to identify anthropomorphic mechanisms in HRI and their effect on human acceptance in manufacturing. Using the PRISMA methodology, a systematic literature review was conducted across the WOS, EBSCO, and SCOPUS databases, resulting in the selection of four articles for final analysis. A quality assessment of the articles was conducted. On a scale of 0 to 16, article scores ranged from 10 to 15, with an average score of 13. The findings indicate that while current research provides valuable insights, it has predominantly focused on conventional anthropomorphic mechanisms from social robotics, such as basic human-like features (e.g., facial expressions, gestures), without exploring more advanced or novel traits. This highlights significant room for further exploration and innovation in industrial settings to enhance user acceptance and interaction. The study underscores the necessity for continued research and development to leverage advanced anthropomorphic designs that can better fulfil the goals of Industry 5.0.

Keywords Systematic literature review · Anthropomorphic mechanisms · Acceptance · Human–robot interaction · Manufacturing context

1 Introduction

The evolution towards Industry 5.0 represents a paradigm shift towards a more sustainable, resilient, and human-centric manufacturing industry. This transition emphasises the need to augment the physical capabilities of robots with cognitive and empathetic qualities, allowing for a deeper integration of human ingenuity and robotic efficiency [1]. The resilience of manufacturing systems, especially highlighted during challenges such as the COVID-19 pandemic, further underscores the importance of adaptive and flexible human–robot collaboration (HRC) systems that can respond dynamically to changing operational demands [2].

Anthropomorphism in robotics, attributing human-like traits to robots, is likely to be significant to their acceptance and integration into manufacturing. This strategy enhances human–robot interaction (HRI) by making interactions more intuitive and increasing safety and efficiency in

✉ Ana Pinto
ana.pinto@dem.uc.pt

¹ Centre for Business and Economics Research (CEBER) and Institute for Systems and Robotic (ISR), Faculty of Sciences and Technology, University of Coimbra, Coimbra, Portugal

² Department of Mechanical Engineering, University of Coimbra, Coimbra, Portugal

³ Institute of Systems and Robotics (ISR), Faculty of Sciences and Technology, University of Coimbra, Coimbra, Portugal

⁴ Centre for Enterprise Systems Engineering, Institute for Systems and Computer Engineering, Technology and Science, Faculty of Engineering, University of Porto, Porto, Portugal

collaborative work. Projects like FourByThree highlight how these human-like robots can improve communication and collaboration, fostering better understanding and teamwork between humans and robots [3]. Moreover, the resilience of manufacturing systems, especially highlighted during challenges such as the COVID-19 pandemic, further underscores the importance of adaptive and flexible HRC systems that can respond dynamically to changing operational demands [1]. The future of manufacturing lies in the seamless integration of these elements, fostering environments where humans and robots can collaboratively innovate and thrive.

There remains a significant knowledge gap concerning the impact of anthropomorphic mechanisms on assembly workers' acceptance of human–robot interaction (HRI). The most relevant work in this domain is the 2024 PRISMA literature review by Puspanathan et al. [4], which briefly references six articles discussing robot-related factors. However, while "trustworthy robot design" emerged as the most frequently discussed factor among the 48 examined in their 14-article sample, the review predominantly focuses on worker-specific acceptance factors, leaving the influence of anthropomorphic mechanisms largely unaddressed.

Industrial HRI acceptance and anthropomorphic mechanisms have been explored separately in the literature. Meissner et al. [5] analysed factors influencing assembly worker acceptance through interviews with 17 German manufacturers. Anthropomorphic mechanisms were examined by Su et al. [6] and Lackey et al. [7]. Expanding beyond industrial robots, Yamazoe et al. [8] investigated how approach kinematics, direction, proxemics, and pauses influenced human impressions of a cleaning rover, indirectly assessing anthropomorphic effects on acceptance. Additionally, literature reviews exist on social robot acceptance [9], humanoid robot acceptance [10], and service robot acceptance by workers [11]. This foundation highlights the need for further study on the role of anthropomorphic mechanisms in assembly worker HRI acceptance.

Integrating anthropomorphic mechanisms in robots represents a vital step towards realising the vision of Industry 5.0, characterised by enhanced HRI, safety, and trust, underpinning a more resilient, sustainable, and human-centric manufacturing environment [1]. Through a systematic literature review, this study review aims to identify anthropomorphic mechanisms in HRI and their effect on human acceptance in the manufacturing context.

In addition to this initial introduction, the organisation of this article unfolds as outlined below: Section 2 delineates the basic concepts associated with human–robot interaction, acceptance and anthropomorphic mechanisms of robots. Section 3 outlines the systematic literature review methodology applied in conducting this study. Section 4 presents the findings of the systematic literature review and elucidates the compilation of guidelines and recommendations gleaned

from the literature for shaping collaborative workspaces conducive to successful interactions between "humans" and "cobots". It also introduces emerging areas for future research. Lastly, Section 6 summarises the conclusions drawn from this research.

2 Basic Concepts

2.1 Human–Robot Interaction

Contrary to the evolution of robotics from the early days to the current state of the art and the predominance of robot types used nowadays, human–robot interaction typically makes one think more of social robots and humanoid-like robots than industrial manipulators. The reason may be that most of us view industrial robots as machines that fulfil repetitive tasks and not as interactive devices. Most industrial applications still have robots "protected" by safety fences that prevent workers from physical contact. This is justified given the accidents that occurred with these powerful machines and that resulted in severe injuries and unfortunate death of workers, who, for one reason or another, were caught inside the robot's operating volume. Although the concept of collaborative industrial robots was proposed in 1996 [12] and after many researchers have proposed control approaches to enable physical interaction with humans, these have only recently paved their way to industry, still following strict regulations that slowly have been softened to allow for the presence of humans in their vicinity, as long as their operations obey speed and force constraints, or if virtual fencing is active to immediately stop their operation if a human enters some established guard zone. Some authors, such as [13], define different collaboration levels, such as coexistence, cooperation, and collaboration.

For coexistence, the robots and humans only share the space without real interaction. On the cooperation level, both participate in some shared goal using a shared space but without necessarily interacting with each other. At the collaboration level, humans and robots share the same goals and act upon shared items, e.g., during a collaborative assembly process.

Nowadays, there is a widespread lack of cobots in industry, despite a push for their integration as part of the Industry 4.0 initiative defined in 2011–13 [14, 15]. This is mainly due to humans' lack of acceptance of them. Enterprises often consider them less useful than cheaper, more specialised industrial robots [14]. Meanwhile, workers perceive them as a threat, whether due to fear of injury, layoffs, or being forced to work at a suboptimal pace [5].

This negative perception may result from overbearing safety standards rather than a genuine lack of aptitude. The behaviour of cobots in HRI scenarios is dictated by ISO

10218–1/2:2011 [16, 17] and ISO/TS 15066 [18]: standards which, despite granting provisions for collaborative operation between humans and robots, allow only excessively constricting modes under which it can take place. Even at their most lenient, they enforce that the robot must reduce speed and/or exerted force while working near a human. In effect, cobots in HRI scenarios can continue performing their assigned tasks, but only poorly, thus exacerbating their reputation for incompetence and hazardousness.

Observing beyond the current state of affairs, the ideal of integrating adaptable robots with the view of benefitting from the niches provided by humans and robots together is a worthwhile one. Firstly, it has been observed that this factory configuration offers the best throughput, even better than lights-out factories [14]. Secondly, industrial workspaces feature tasks that humans are not suited for, whether due to possible environmental hazards, health risks, or fatigue-inducing repetition, and robots are ideally equipped to relieve our burden in those cases. Finally, though the increased robotisation of industrial workspaces will likely continue, the HRI goals of Industry 5.0 offer a way forward that secures a worthwhile role for humans in factories and, at the same time, ensures optimal profitability, thus providing a motivator that is not merely civic.

2.2 Acceptance

A subject's acceptance of an object relates to their approval of the outcomes caused by its presence [5]. In robotics, because the concept had initially sparked interest specifically for the cases of social and service robots, a narrower definition of human acceptance toward robots has historically been employed, that it represents a subject's willingness to make use of the robot [19]: what has otherwise been termed "behavioural acceptance" [20]. In light of this concept understanding, the standard adopted experimental acceptance measures have been based on a human subject's choice to use a target robot, namely 21's Technology Acceptance Model [21] and its derivatives, like the Almere model [22]. These measures accurately represent the subject's acceptance in cases where they can use a robot, which is usually valid for social and service robots. However, the same measures have also been widely applied to industrial HRI, where a worker's ability to choose a robot is not guaranteed. However, an increasing number of authors have called attention to this discrepancy [5, 23]. These authors argue for the measurement instead of what has been termed "attitudinal acceptance" [20]: the impression or assessment that a subject has for an object, most directly obtainable through qualitative measures such as interviews and questionnaires. However, no widely accepted standard questionnaire for industrial robot acceptance exists.

The factors influencing acceptance are threefold: beyond the characteristics of the person and the robot, the context within which the industrial HRI occurs is also an influence. For example, workers participating in commissioning and integrating a robot will typically experience an increased acceptance of its presence in the workspace [5, 14]. For the scope of this work, however, only factors related to the robot itself should be considered, as those are the ones that can be affected by the implementation of anthropomorphic mechanisms.

Based on a qualitative analysis performed on the results of 17 interviews with workers bearing various levels of HRC experience, Meissner et al. [5] divided robot-related acceptance factors into six categories: safety; process reliability (i.e. task competency); adjustability to workers; adjustability to tasks and environments; trustworthy design (i.e. shape and behaviour exuding good intentions and affordances); and controllable and autonomous workflow (i.e. acceptance increases if the robot stays in a subordinate role).

Of those categories, the one where anthropomorphic mechanisms have the most potential to produce a positive impact is trustworthy design, specifically in shaping the robot's behaviour. For example, a robot exhibiting anthropomorphic characteristics in its behaviour (as opposed to merely performing its intended action) can more effectively allow the human to glean its plans, what it can do, and what actions it would never take. In turn, remaining consistent with human expectations will likely correlate to increased trust.

2.3 Anthropomorphic Mechanisms of Robots

Within this literature review, anthropomorphic mechanisms were considered any that a robot may use to render the interaction more human-readable, hence the term "anthropomorphic". This naturally encompassed mechanisms for expression (i.e., social cues) but also included mechanisms for recognition, as they were taken to also indirectly render the interaction more predictable. Regarding expression, the mechanisms could be human- or animal-like (e.g. speech, gestures, body language, etc.) or have no biological analogue (e.g. a display scrolling text).

Su et al. [6] and Lackey et al. [7] have produced studies into multimodal HRI that provide the most complete picture of proposed anthropomorphic mechanisms, compiled in Table 1. Explicit mechanisms are associated in human-to-human communication with purposeful cues with defined meanings. In contrast, implicit mechanisms would emulate implicit human-to-human communication: modalities where emotional or contextual state information is inadvertently delivered, thereby influencing the interpretation of associated explicit communication [7].

Table 1 Table of anthropomorphic mechanisms for expression available to robots, sorted by their potential applicability to industrial HRI according to typical cobot designs. A reason for applicability is provided for each mechanism as well

Anthropomorphic mechanism for expression	Industrial HRI applicability	Reason for applicability	Target sense	Implicit or Explicit	Depends on	Ref
Gestures (pointing, hand signals)	High	Arm can be used to gesticulate	Sight	Explicit	-	[6] [7]
Lights (constant, flashing, pulsating, etc.)	High	Easy to implement; Easily visible	Sight	Explicit	-	[6]
Alarm sounds	High	Although factory settings are noisy, unoccupied frequency ranges can still be found; Klaxons can easily be quite loud	Hearing	Explicit	-	[7]
Force feedback	High	Can be used to convey levels of resistance to certain actions during teaching	Touch	Explicit	-	[6] [7]
Gaze direction	High	Pointing the end-effector or a camera can be used as a surrogate for gaze	Sight	Implicit	-	[6] [7]
Body language	High	The manner in which gestures are performed (gradually, frantically, etc.) is easily visible	Sight	Implicit	Gestures	[6] [7]
Posture	High	Level of alertness can be conveyed by how tense / relaxed the arm is	Sight	Implicit	-	[7]
Proxemics	High	Applicable to both stationary and (especially) mobile robots	Sight	Implicit	-	[7]
Co-verbal gestures (talking with hands)	High	Arm can be used to gesticulate	Sight	Implicit	Speech	[6] [7]
Speech	Medium	Speech can be used to convey information at a very close distance	Hearing	Explicit	Message semantics	[6] [7]
Touch patterns	Medium	Light touching (e.g. tapping on shoulder) can be used to call attention, though in most cases the robot and human should not touch at all	Touch	Explicit	-	[6] [7]
Gait	Medium	Applicable to mobile robots, in the form of travel speed	Sight	Implicit	-	[6] [7]
Touch emphasis (force pattern, shakiness, etc.)	Medium	Can be used to reinforce a cue given by touch patterns, but requires touching, which is inadvisable	Touch	Implicit	Touch patterns	[7]
Message semantics	Medium	Speech can be used to transmit information	Hearing / Sight	Explicit	-	-
Display messages (chatbox)	Low	Screens on industrial robots in shared tasks are inadvisable: they distract from hazards	Sight	Explicit	Speech	[6]
Object / space displays, high-lighted details	Low	Screens on industrial robots in shared tasks are inadvisable: they distract from hazards	Sight	Explicit	Message semantics	[6]
Facial expression (display, moving parts)	Low	Faces on industrial robots in shared tasks are inadvisable: they distract from hazards	Sight	Implicit	-	[6] [7]
Speech expressiveness (tonality, rate, intensity)	Low	Difficult to notice subtle details in speech in a noisy environment	Hearing	Implicit	Speech	[7]

Mechanisms were sorted according to their potential for applicability to industrial HRI, based on the components that a cobot will either usually have (an arm with

end-effector, no screen, etc.) or can easily be attached to it at low cost (lights, klaxons, etc.). The result is that, for example, anthropomorphic mechanisms related to

movement (gestures, posture, body language) were deemed very applicable to industrial robots. Virtual fencing and the expected future prominence [14] of mobile cobots (e.g. Rob@Work 3 [25]) render proxemics viable for industrial HRI as well. Speech was considered not very feasible on account of it not being reliable as a mode of communication in a factory setting beyond a distance of 3 m [24] (or even less in especially noisy environments), and mechanisms related to screen displays were considered inadvisable on account of the danger caused by distracting the worker from potential hazards [19]. Further explanations are provided in the table itself.

3 Systematic Literature Review Method

This systematic literature review followed the steps proposed by Donato and Donato [26]. According to these authors, systematic reviews set themselves apart from traditional ones by being both replicable and impartial. The review aimed to be comprehensive, encompassing all pertinent literature and adhering to a stringent [27] methodology. The initial phase involved crafting the research questions. Once the research focus for the review was clearly defined, the inclusion and exclusion criteria and the search strategy were established. Subsequently, articles were chosen. Finally, data was extracted and synthetically analysed. The methodological details of these stages are outlined in the subsequent sections, following the PRISMA framework [27]. The PICO structure (Population or Problem, Interest, Context, Outcome) was employed to formulate the research questions. The specific research question formulated was: What anthropomorphic mechanisms exist in

HRI (in an industrial environment), and what is their effect on human acceptance?

3.1 Eligibility Criteria

This review includes literature that was written in English. Only articles or conference articles were included. The studies that were not developed in an industrial context or did not fit the theme (i.e., were not actually in the scope of human–robot interaction, robots' anthropomorphic mechanisms and mention of user acceptance) were excluded. The remaining studies were analysed by reading the full text.

3.2 Search Strategy

The exploration approach encompassed a thorough search designed to identify a broad range of articles for thorough examination. This search was conducted across specific electronic databases, namely Web of Science (all databases), EBSCO and SCOPUS. These databases were selected due to their extensive relevance to the study's topics and widespread utilisation within the field. The keywords employed in this literature review included anthropomorphic mechanisms, acceptance, human–robot interaction, robots, manufacturing context, and users. Each of these keywords was paired with its corresponding similar words, as illustrated in Table 2.

3.3 Screening Criteria

No temporal constraints were imposed on including publications; the articles were collected on December 3, 2023. The results retrieval unfolded in two distinctive steps after the search phase. Initially, the articles were inputted into an electronic spreadsheet, and any instances of duplicated

Table 2 Keywords and their similar words

Keyword	Similar words	Keyword	Similar words
Anthropomorphic mechanisms	Human* mechanisms Anthropo* mechanisms Anthropo* attributes	Robots	Robo* Machine Automaton Bot
Acceptance	Accept*	Manufacturing context	Producti* Industr* Lab* Manufactur* Factor* Plant
Human Robot Interaction	human–robot interac* human robot interact* Human-interact* robot HRI Human Machine Interaction Collab* robot Robot Human Interface	Users	Laborer* Operator* Human* Worker User*

studies were expunged. Subsequently, adhering to the PRISMA framework, a comprehensive examination was carried out, encompassing the scrutiny of titles, abstracts, and keywords for all remaining articles. The criteria for progression to the subsequent phase dictated that the articles must encompass discussions on human–robot interaction, delve into anthropomorphic mechanisms in robots, and touch upon user acceptance. When the fulfilment of these criteria was unclear, the articles were included for further consideration.

3.4 Quality and Bias Metrics

The study's quality was evaluated using criteria tailored to the review's objective, as outlined by Law et al. [28]. Table 3 provides the specific quality criteria and their definitions. Two reviewers independently assessed the quality of the articles. Quality scores were determined by rating each criterion as presented (yes = 1 point), not given (no = 0 points), or not applicable (NA).

Table 3 Definition and description of the quality assessment and risk of bias methodology

Quality assessment criteria	
Q1	Was the study purpose stated clearly?
Q2	Was relevant background literature reviewed?
Q3	Was the design appropriate for the research question?
Q4	Was the sample described in detail?
Q5	Was sample size justified?
Q6	Was informed consent obtained?
Q7	Were the outcome measures reliable?
Q8	Were the outcome measures valid?
Q9	Was the method described in detail?
Q10	Were results reported in terms of statistical significance?
Q11	Were the analysis methods appropriate?
Q12	Was importance for the practice reported?
Q13	Were any dropouts reported?
Q14	Were conclusions appropriate given the study methods?
Q15	Are there any implications for practice given the results of the study?
Q16	Were limitations of the study acknowledged and described by the authors?

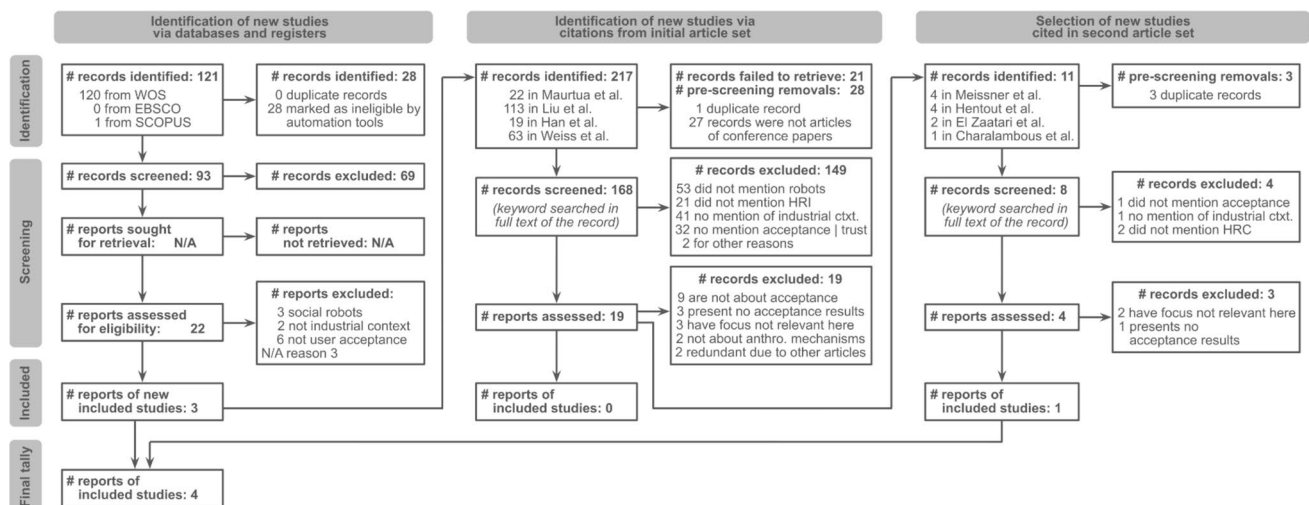


Fig. 1 Flow diagram for the PRISMA procedure applied to references

4 Results

4.1 Study Identification and Selection

The flow diagram of the PRISMA methodology is presented in Fig. 1. Initially (leftmost column), 121 records were obtained (no duplicates). Automated tools automatically excluded twenty-eight articles, leaving us with 93 articles. Of these, 69 articles were excluded based on reading the title, abstract, and keywords. We were left with 22 articles. Upon full reading, four of these 22 articles were about social robots, 11 did not consider user acceptance, one did not discuss anthropomorphic mechanisms, and two were not conducted in an industrial context. Thus, we ended up with three articles that addressed our research question.

To validate the state-of-the-art representativity of the obtained batch, a second PRISMA pass (centre column of Fig. 1) was performed with the references from the three articles initially obtained, in addition to another one (Weiss et al. [29]) that was initially considered for inclusion but eventually rejected due to lack of focus on anthropomorphic mechanisms during the data analysis phase in the article itself. However, it is a part of a larger project (AssistMe) involving anthropomorphic mechanisms. Two hundred sixteen distinct references exist across them, not counting one duplicate. It was impossible to retrieve 21 records; another 27 were excluded immediately because they were not journal articles or conference articles. The remaining 168 records were filtered for the presence of the strings “robo*”; “indus*”; “human–robot interaction” or “HRI” in the article text, eliminating another 115. Another 32 were filtered out for missing reference to acceptance, and another two for other reasons (one only referred to industrial robotics in passing; the other was a set of recommendations for implementing the Industry 4.0 initiative but did not provide experimental data). Upon reading the remaining 19 articles, it was decided that none were eligible for this review. Still, four contained references to potentially eligible articles, at least regarding describing

their respective results. A final PRISMA pass was performed on those.

The flow diagram of the final PRISMA pass is presented in the right column of Fig. 1. Excluding duplicates, eight records were obtained. The same filtering procedure was performed as in the second PRISMA pass during the screening step, with four articles.

excluded for missing keywords in the text. Of the remaining articles, two focused on aspects besides acceptance and anthropomorphic mechanisms despite addressing industrial HRC, and one did not present any acceptance results, merely proposing an architecture joining together prior works (already referenced). The article, due to Elprama et al. [23], was included into our final sample. All four selected articles are listed in Table 4.

4.2 Quality Assessment

The metrics for the journals/conference articles of each study are presented in Table 4. Of the four articles analysed, 1 is Quartile 1, another is Quartile 2, and the other two have no Quartile defined. The data refers to 2023.

The quality scores for the articles are presented in Table 5. On a scale of 0 to 16, article scores ranged from 10 to 15. Liu et al. (2023) and Elbrama et al. (2016) had the highest quality score, and Maurtura et al. (2017) had the lowest. The average score was 13, with the following distribution: 2 articles scored 15 points, one article scored 12 points, and one article scored 10 points. Regarding the quality assessment criteria, Q1, Q2, Q3, Q4, Q7, Q9, Q12, Q14 and Q15 were fulfilled in all articles. Q5 was fulfilled by two articles, Q6 by three articles, Q8 by two articles, Q10 by three articles, Q11 by three articles, and Q16 by three articles.

4.3 Data Analysis

This study aims to characterise the sample by analysing the descriptive results from four experimental articles, encompassing laboratory and real-world settings. The distribution of these articles spans several years, with one

Table 4 Metrics for the journal/conference article

Ref	Journal / Conference Paper	Publisher	Quartile	Impact Factor
Liu et al.[19]	International Journal of Human–Computer Interaction	Taylor & Fancis	Q1	5.8
Maurtua et al.[3]Study 1 + Study 2	International Journal of Advanced Robotic Systems	Sage	Q2	2.3
Han et al.[30]	Procedia Manufacturing*	Elsevier Inc	—	—
Elprama et al.[23]	25th IEEE International Symposium on Robot and Human Interactive Communication	IEEE	—	—

*"Procedia Manufacturing" has been discontinued since 2022

Table 5 Articles quality scores

Reference	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	—
Elprama et al.[23]	1	1	1	1	1	1	1	1	—
Han et al.[30]	1	1	1	1	0	1	1	0	—
Liu et al.[19]	1	1	1	1	1	1	1	1	—
Maurtua et al.[3]	1	1	1	1	0	0	1	0	—
Study 1 + Study 2									
Reference	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Score
Elprama et al.[23]	1	1	1	1	0	1	1	1	15
Han et al.[30]	1	1	1	1	0	1	1	0	12
Liu et al.[19]	1	1	1	1	0	1	1	1	15
Maurtua et al.[3]	1	0	0	1	0	1	1	1	10
Study 1 + Study 2									

Table 6 Articles considered for analysis. The method column characterises the application context: either experimental in lab (E.L.); or experimental in a real-world environment (E.R.)

Authors	Title	Type of article	Year	Sample	Method
Maurtua et al. [3] Study 1 + Study 2	Human–robot collaboration in industrial applications: Safety, interaction and trust	Article	2017	38 participants in study 1 and 77 participants in study 2	E.L
Liu et al. [19]	The Effects of Type and Form of Collaborative Robots in Manufacturing on Trustworthiness, Risk Perceived, and Acceptance	Article	2023	40 participants	E.L
Han et al. [30]	Cognitive Human–Robot–Collaboration in Assembly: A framework for Cognitive Interaction Planning and Subject Study	Proceeding Article	2021	60 participants of BMW	E.R
Elprama et al. [23]	Acceptance of collaborative robots by factory workers: a pilot study on the importance of social cues of anthropomorphic robots	Proceeding Article	2016	11 employees working in the body shop of a car factory	E.L

published in 2016, 2017, 2021, and 2023 respectively. This timeline of publications includes two journal articles and two conference articles, highlighting a diverse array of dissemination platforms and academic forums. The experimental nature of these articles is divided into three conducted purely in laboratory environments, and one in a real-world setting, thereby providing a good framework for evaluating the underlying research phenomena. Table 6 briefly describes the articles considered for analysis, including information on the authors, title, type of article, year of publication, sample of participants, and method used.

Table 7 lists qualitative descriptions of the cobots used in each study (Maurtua et al. was divided into the two distinct studies described in the same article, and it should be noted that Liu et al. only used video footage); the cobot form and shape (i.e. presence of screen and number of arms); anthropomorphic mechanisms; the method of interaction that the human had with the robot during the experiment; the independent variable (when applicable: several studies did not change any parameter to produce their results); and

the observed result on the participants' acceptance toward the robot(s).

Regarding cobot models, the most common were Universal Robots UR10 and Rethink Robotics Baxter, both featuring twice. KUKA iiwa, ABB YUMI-IRB 14000 and 14,050, and Rethink Robotics Sawyer all featured once. It should also be reiterated here that the obtained batch of articles is due to three rounds of PRISMA search. At this point, potentially eligible encountered records began to be most often ones already included previously, so it is unlikely that any significant articles have been missed.

By comparing the anthropomorphic mechanisms tested in Table 7 to the potentially applicable ones from Table 1, it is notable how underexplored the subject is within industrial HRI. The number of present studies is very small, and anthropomorphic mechanisms were only properly explored in half of them.

The only study to extend beyond typical social robotics anthropomorphic mechanisms was Maurtua et al.'s Study 2, where a robot was fitted with an LED strip that would light up in acknowledgement that a human-issued command had

Table 7 Table listing cobot models and their features, the nature of the experiment, and the observed effect on acceptance

Ref	Cobot	Screen and Arms	Anthro. Mechanisms	Interaction	Independent Variable	Effect on Acceptance
Elprama et al. [23]	Baxter	Yes 2	- Nodding; - Eyes shown; - Turn “head” to target	Instructing	Less vs. More social expression	Increased with social expression
Maurtua et al. [3] Study 1	UR10	No 1	Gesture recognition	Instructing and interfering with robot’s personal space	N/A	Accepted
Maurtua et al. [3] Study 2	iiwa	No 1	LED strip for visual feedback	Instructing	N/A	Increased perceived usefulness and responsiveness vs. Study 1
Han et al. [30]	UR10	No 1	- Intention estimation; - Obstacle avoidance	Turn-taking on shared workbench	Recognition modalities	Slight increase with recognition modalities
Liu et al. [19]	Yumi	No 2	None	Video footage of each robot performing pick-and-place task	- Number of arms; - Presence of screen	No effect
	Yumi	No 1	None			
	Baxter	Yes 2	Eyes shown			
	Sawyer	Yes 1	Eyes shown			

been received [3]. It is notable how such a relatively simple feature significantly improved participants’ acceptance of the robot, especially when contrasted with Han et al.’s implementation [30], which had an elaborate and very effective architecture for intention estimation, path planning, real-time obstacle avoidance and speed adjustment, but only yielded mild improvements in acceptance, despite excellent results otherwise [30]. The strength of this example is further compounded by the acceptance level in Maurtua et al. Study 1 and Han et al.’s control test being reasonably comparable.

Finally, Liu et al.’s study is an outlier relative to the rest of the batch. It differs significantly in terms of method, and the independent variable was not an anthropomorphic mechanism but instead the *appearance* of the cobot [19]. Still, the features studied applied to the design of an industrial robot. Although no statistically significant effect was observed on acceptance due to the presence of a screen and the number of arms, it was advised by the authors that a screen for facial display may not be suitable in an industrial setting on account of the additional distraction generated.

5 Avenues for Future Research

In the context of Industry 5.0, which represents the integration of advanced technologies, including robotics, into industrial processes, the role of anthropomorphic mechanisms becomes critical. As industries evolve towards greater automation and connectivity, robots with anthropomorphic features can contribute to a more seamless collaboration between human workers and robotic systems. This collaborative approach aims to combine the strengths of both

humans and robots, fostering a more flexible and efficient production environment.

One should remember that the primary reason for integrating robots into industry is their flexibility and adaptability to ever-changing production requirements. Distinct from traditional industrial pipelines that focus on the mass production of a single product, modern production lines need to be versatile enough to handle a constantly changing variety of products. Robots are invaluable for this purpose, but this adaptability often necessitates the inclusion of “humans-in-the-loop,” highlighting the importance of human–robot collaboration.

When performing repetitive tasks, humans tend to develop automatic behaviours and become accustomed to the operation of nearby machines and robots. However, with current production trends requiring frequent changes in robot operation for different products, human operators face new challenges. They need to understand and predict robot movements to maintain productivity and ensure safety, both critical for an efficient production environment.

Therefore, the field of human–robot collaboration presents significant research opportunities. Future research should explore a broader range of anthropomorphic mechanisms and their impacts on various aspects of HRI, including long-term acceptance, safety, and productivity.

Investigating which anthropomorphic traits in robots—such as organic and intuitive movements—can enhance human comfort and acceptance is crucial, including long-term acceptance, safety, and productivity. Another area of research is the adaptation of the robot behaviours based on an understanding of the human operator tasks and even on their focus of attention or emotional state [31]. Understanding these dynamics can lead to the development of robots

that not only increase productivity but also ensure a safer working environment.

Additionally, research can focus on optimising robot motion patterns, improving human–robot communication interfaces, and studying the psychological impacts of working alongside robots exhibiting anthropomorphic characteristics and behaviours. Control theory approaches such as those explored in [32–36] may aid with these future directions. These studies can inform the design of future robotic systems that are better integrated into human-centric production environments, ultimately driving advancements in Industry 5.0. Furthermore, exploring socio-psychological factors that influence worker acceptance of robots could provide deeper insights into designing more effective interactive robotic systems. Finally, research into industry-centric anthropomorphic mechanisms may have applications beyond this scope, for example in honing the expressive deftness of cobots applied to art production, such as is the case of the e-David, Karimov et al., and FRIDA robotic painters [37–39]. By addressing these research areas, the field can advance towards creating robots that enhance human well-being and efficiency in industrial settings.

In practice, a wide span is possible for future research avenues: as demonstrated in this review, the effect of anthropomorphic mechanisms on industrial HRI acceptance is currently poorly understood. There are indications that their use is critical to worker acceptance, yet most of the viable anthropomorphic mechanisms have remained underexplored for this purpose. Thus, a possible roadmap would have to cycle between implementing individual anthropomorphic mechanisms and testing their effect on acceptance. Implementation may begin with the ones bearing high industrial HRI applicability whilst also being easy, be it due to an inherent lack of complexity (e.g. adding lights) or due to existing precedents from service or social robotics. Then, it may progress in descending order of ease and applicability. Testing may be achieved using Wizard-of-Oz experiments and may be benchmarked against non-expressive cobots.

6 Conclusions

This research aims to identify anthropomorphic mechanisms in HRI and their effect on human acceptance in manufacturing, conducted through a systematic literature review.

The findings of this study underscore the significant role that anthropomorphic mechanisms play in enhancing the acceptance of robots in manufacturing settings. Including human-like features, such as facial expressions, voice recognition, gestures, and social behaviours, may improve communication between humans and robots, thereby increasing acceptance and trust among workers, and the study performed here displays encouraging evidence supporting this view. Recognition-based

features facilitate more intuitive and efficient interaction comprehension, showing potential for making collaborative tasks smoother and safer, though the effect on acceptance of such features was minor. Expressive features, adjusting the behaviour of the robot for better readability, were seen to significantly influence user acceptance in our sample, with robots that convey their intentions and capabilities being more readily accepted by human workers. Furthermore, the robot's visual design was also observed to be an important factor.

Despite the recognised importance of these mechanisms, the literature on this subject is sparse. Only a few studies have thoroughly explored the potential of anthropomorphic mechanisms in industrial HRI. This research consolidates existing knowledge, identifies critical gaps, and highlights the need for further investigation into the design and implementation of anthropomorphic features in industrial robots. The limited number of articles included in the analysis—only four out of 324—poses a significant limitation. It should be reiterated that the four selected articles (the studies selected had a ten or higher quality score) are the outcome of performing three rounds of replicable new study identifications, with the last one yielding more duplicates than new documents, so while it is always possible that some relevant studies may have been missed due to keyword selection, database limitations, or publication biases, the comprehensiveness of the systematic review method poses compelling evidence for said sparseness.

Important conclusions drawn from this research include the need for increased investment in developing anthropomorphic mechanisms in industrial robots. Clear guidelines for designing and implementing these features should be developed to ensure their effective integration into manufacturing environments. Future research should explore a broader range of anthropomorphic mechanisms and their impacts on various aspects of HRI, including long-term acceptance, safety, and productivity. Additionally, investigating the socio-psychological factors that influence worker acceptance of robots could provide deeper insights into designing more effective interactive robotic systems.

From a practical perspective, the findings suggest that industries should consider adopting robots with anthropomorphic features to improve worker acceptance, satisfaction, safety, and overall productivity. This approach aligns with the goals of Industry 5.0, which aims for a more human-centric and resilient manufacturing environment.

In conclusion, this systematic literature review emphasises the critical role of anthropomorphic mechanisms in enhancing user acceptance of robots in manufacturing settings. Although the current literature is limited, the findings underscore the potential benefits of human-like robots in fostering effective human–robot collaboration. This research paves the way for more advanced and integrated industrial systems, highlighting the need for ongoing research and development to fully realise the potential of Industry 5.0.

Authors' Contribution The idea for the article by Ana Luisa Sousa Pinto. All authors contributed to the literature search, and Ana Luisa Sousa Pinto and Andrey Solovov performed data analysis. All authors wrote the first draft of the manuscript, and it was then critically revised by Ana Luisa Sousa Pinto, Ana Cristina Correia Simões, and Paulo Jorge Carvalho Menezes. All authors read and approved the final manuscript.

Funding Open access funding provided by FCTIFCCN (b-on). This work has been funded by the FCT/MCTES projects UIDB/00048/2020 (DOI: 10.54499/UIDB/00048/2020) and UIDB/05037/2020 (DOI: 10.54499/UIDB/05037/2020); by the PRR—Recovery and Resilience Plan and the Next Generation EU European Funds, following NOTICE No. 02/C05-i01/2022, Component 5—Capitalization and Business Innovation—Mobilizing Agendas for Business Innovation, under the project Greenauto—PPS10/PPS12/PPS13, reference 7255, C629367795-00464440.

Data Availability The data generated and analysed as part of this systematic review is available in the Zenodo repository, <https://zenodo.org/doi/10.5281/zenodo.12663097>

Declarations

Competing Interests The authors have no employment interests, financial interests, professional interests or other competing interests to declare that are relevant to the content of this article.

Ethics Approval Not applicable.

Consent to Publish Not applicable.

Consent to Participate Not applicable.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Singh, T., Singh, D., Singh, C.D., Singh, K.: Industry 5.0. In: Singh, C.D., Kaur, H. (eds.) *Factories of the Future*, pp. 21–45. Scrivener Publishing LLC (2023). <https://doi.org/10.1002/9781119865216.ch2>
- Romero, D., Stahre, J.: Towards the resilient operator 5.0: the future of work in smart resilient manufacturing systems. *Procedia CIRP* (2021). <https://doi.org/10.1016/j.procir.2021.11.183>
- Maurtua, I., Ibarguren, A., Kildal, J., Susperregi, L., Sierra, B.: Human–robot collaboration in industrial applications: safety, interaction and trust. *Int. J. Adv. Robot. Syst* (2017). <https://doi.org/10.1177/1729881417716010>
- Puspanathan, T., Bitsch, G., Louw, L.: Systematic literature review on the acceptance of human-robot collaboration among assembly workers. *Procedia CIRP* **126**, 188–193 (2024)
- Meissner, A., Trübswetter, A., Conti-Kufner, A.S., Schmidler, J.: Friend or foe? Understanding assembly workers' acceptance of human-robot collaboration. *ACM Trans. HRI*. (2020). <https://doi.org/10.1145/3399433>
- Su, H., Qi, W., Chen, J., Yang, C., Sandoval, J., Laribi, M.A.: Recent advancements in multimodal human–robot interaction. *Front. Neurobot.* (2023). <https://doi.org/10.3389/fnbot.2023.1084000>
- Lackey, S., Barber, D., Reinerman, L., Badler, N.I., Hudson, I.: Defining next-generation multi-modal communication in human robot interaction. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting* (2011). <https://doi.org/10.1177/1071181311551095>
- Yamazoe, H. et al. Analysis of impressions of robot by changing its motion and trajectory parameters for designing parameterized behaviors of home-service robots. *Intel Serv Robotics* (2022) <https://doi.org/10.1007/s11370-022-00447-1>
- David, D., Thérouanne, P., Milhabet, I.: The acceptability of social robots: A systematic review of the recent literature. *Comput. Hum. Behav.* **137**, 107419 (2022)
- Moberg, R., Khan, A.: Humanoid robot acceptance: a concise review of literature. in *2022 International Conference on Computational Science and Computational Intelligence (CSCI) 1223–1228* (IEEE, Las Vegas, NV, USA, 2022). <https://doi.org/10.1109/CSCI58124.2022.00219>
- Xu, J., Hsiao, A., Reid, S., Ma, E.: Working with service robots? A systematic literature review of hospitality employees' perspectives. *Int. J. Hosp. Manag.* **113**, 103523 (2023)
- Colgate, J.E., Wannasuphprasit, W., Peshkin, M.A.: Cobots: robots for collaboration with human operators. *Proceedings of the International Mechanical Engineering Congress and Exhibition*, Atlanta, GA (1996). <https://doi.org/10.1115/IMECE1996-0367>
- Aaltonen I., Salmi T., Marstio I.: Refining levels of collaboration to support the design and evaluation of human-robot interaction in the manufacturing industry. *Procedia CIRP* (2018). <https://doi.org/10.1016/j.procir.2018.03.214>
- Sanneman, L., Fourie, C., Julie, S.A.: The state of industrial robotics: emerging technologies, challenges, and key research directions. *now* (2021). <https://doi.org/10.48550/ARXIV.2010.14537>
- Kagermann, H., Wahlster, W., Helbig, J.: Recommendations for implementing the strategic initiative INDUSTRIE 4.0 -- securing the future of german manufacturing industry. <https://en.acatech.de/publication/recommendations-for-implementing-the-strategic-initiative-industrie-4-0-final-report-of-the-industrie-4-0-working-group/> (2013). Accessed 1 May 2024
- International Organization for Standardization: ISO 10218–1:2011 – Robots and robotic devices – Safety requirements for industrial robots – Part 1: Robots. <https://www.iso.org/standard/51330.html> (2011). Accessed 1 May 2024
- International Organization for Standardization: ISO 10218–2:2011 – Robots and robotic devices – Safety requirements for industrial robots – Part 2: Robot systems and integration. <https://www.iso.org/standard/41571.html> (2011). Accessed 1 May 2024
- International Organization for Standardization: ISO/TS 15066:2016 – Robots and robotic devices – Collaborative robots. <https://www.iso.org/standard/62996.html> (2016). Accessed 1 May 2024
- Liu, L., Zou, Z., Greene, R.L.: The effects of type and form of collaborative robots in manufacturing on trustworthiness, risk perceived, and acceptance. *Int. J. HCI* (2023). <https://doi.org/10.1080/10447318.2023.2169527>
- Schäfer, M., Keppler, D.: Modelle der technikorientierten Akzeptanzforschung : Überblick und Reflexion am Beispiel eines Forschungsprojekts zur Implementierung innovativer technischer

- Energieeffizienz-Maßnahmen. <https://depositonce.tu-berlin.de/items/25ac9084-59f3-4231-83a1-a8ffccb8fd1c> (2013). Accessed 1 May 2024
21. Davis, F. D.: A technology acceptance model for empirically testing new end-user information systems : theory and results. <http://hdl.handle.net/1721.1/15192> (1985) Accessed 1 May 2024
 22. Heerink, M., Kröse, B., Evers, V., Wielinga, B.: Assessing acceptance of assistive social agent technology by older adults: the Almere model. *Int. J. Soc. Robot.* (2010). <https://doi.org/10.1007/s12369-010-0068-5>
 23. Elprama, S.A., El Makrini, I., Vanderborcht, B., Jacobs, A.: Acceptance of collaborative robots by factory workers: a pilot study on the importance of social cues of anthropomorphic robots. The 25th IEEE International Symposium on Robot and Human Interactive Communication. <https://researchportal.vub.be/en/publications/acceptance-of-collaborative-robots-by-factory-workers-a-pilot-study> (2016). Accessed 1 May 2024
 24. Barattini, P., Morand, C., Robertson, N.M.: A proposed gesture set for the control of industrial collaborative robots. 2012 IEEE RO-MAN: The 21st IEEE International Symposium on Robot and Human Interactive Communication (2012). <https://doi.org/10.1109/ROMAN.2012.6343743>
 25. Fraunhofer IPA: rob@work 3. <https://www.care-o-bot.de/en/rob-work.html> (2023). Accessed 1 May 2024
 26. Donato, H., Donato, M.: Stages for undertaking a systematic review. *Acta Méd. Port.* (2019). <https://doi.org/10.20344/amp.11923>
 27. Page, M.J., McKenzie, J.E., Bossuyt, P.M., Boutron, I., Hoffman, T.C., Mulrow, C.D., Shamseer, L., Tetzlaff, J.M., Akl, E.A., Brennan, S.E., Chou, R., Glanville, J., Grimshaw, J.M., Hróbjartsson, A., Lalu, M.M., Li, T., Loder, E.W., Mayo-Wilson, E., McDonald, S., McGuinness, L.A., Stewart, L.A., Thomas, J., Tricco, A.C., Welch, V.A., Whiting, P., Moher, D.: The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* (2021). <https://doi.org/10.1136/bmj.n71>
 28. Law, M., Stewart, D., Pollock, M., Letts, L., Bosch, J., & Westmorland, M.: Guidelines for Critical Review Form: Quantitative Studies (1998)
 29. Weiss, A., Huber, A., Minichberger, J., Ikeda, M.: First Application of Robot Teaching in an Existing Industry 4.0 Environment: Does It Really Work? Societies (Basel) (2016). <https://doi.org/10.3390/soc6030020>
 30. Han, L., Yang, S., Tyroller, Q.: Cognitive Human-Robot-Collaboration in Assembly: A Framework for Cognitive Interaction Planning and Subject Study. *Procedia Manuf.* (2021) <https://doi.org/10.1016/j.promfg.2021.10.005>
 31. Zarei, M., Solovov, A., Gautam, V., Ferreira, B., Assunção, G., Marin-Hernandez, A., Menezes, P.: Transforming Collaboration: A Vision for Human-Aware Robots to Enhance Worker Safety and Boost Production Efficiency, IEEE MELECON 2024 - The 22nd IEEE Mediterranean Electrotechnical Conference. Porto, Portugal (2024)
 32. Peng, Z., Song, X., Song, S., Stojanovic, V.: Hysteresis quantified control for switched reaction–diffusion systems and its application. *Complex Intell. Syst.* **9**, 7451–7460 (2023)
 33. Song, X., Peng, Z., Song, S., Stojanovic, V.: Interval observer design for unobservable switched nonlinear partial differential equation systems and its application. *Intl. J. Robust. Nonlinear* **34**, 10990–11009 (2024)
 34. Tao, Y., Tao, H., Zhuang, Z., Stojanovic, V., Paszke, W.: Quantized iterative learning control of communication-constrained systems with encoding and decoding mechanism. *Trans. Inst. Meas. Control* **46**, 1943–1954 (2024)
 35. Tao, H., Zheng, J., Wei, J., Paszke, W., Rogers, E., Stojanovic, V.: Repetitive process based indirect-type iterative learning control for batch processes with model uncertainty and input delay. *J. Process. Control.* **132**, 103112 (2023)
 36. Song, X., Zheng, D., Song, S., Stojanovic, V., Tejado, I.: Robust anti-disturbance interval type-2 fuzzy control for interconnected nonlinear PDE systems via conjunct observer. *Math. Comput. Simul.* **227**, 149–167 (2025)
 37. Gülzow, J.M., Deussen, O.: Region-Based Approaches in Robotic Painting. *Arts* **11**, 77 (2022)
 38. Karimov, A., Kopets, E., Leonov, S., Scalera, L., Butusov, D.: A Robot for Artistic Painting in Authentic Colors. *J. Intell. Robot. Syst.* **107**, 34 (2023)
 39. Schaldenbrand, P., McCann, J., Oh, J.: FRIDA: A Collaborative Robot Painter with a Differentiable, Real2Sim2Real Planning Environment. in 2023 IEEE International Conference on Robotics and Automation (ICRA) 11712–11718 (IEEE, London, United Kingdom, 2023). <https://doi.org/10.1109/ICRA48891.2023.10160702>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Ana Pinto holds a PhD in Organizational, Work, and Human Resources Psychology from the University of Coimbra. She is a faculty member at the Faculty of Science and Technology, a researcher at the Centre for Business and Economics Research (CeBER), and a guest researcher at the Institute of Systems and Robotics (ISR), also at the University of Coimbra. Her main research interests include human-robot interaction, psychosocial risks, work teams, emotional regulation, well-being, and quality of life at work.

Andrey Solovov (Member, IEEE) completed in 2021 a MSc. in astrophysics and space instrumentation at the University of Coimbra (UC), Portugal, working on ML-based rare particle detection. Prior positions have been as Research Fellow for a 19th century astronomical history joint project by the Universities of Coimbra and Aveiro, Portugal; and later the role of Research Engineer, developing a software-as-a-service audiology architecture under the Immersive Systems and Sensory Stimulation Laboratory (IS3L) at the Institute of Systems and Robotics (ISR), Coimbra. His current position is as Research Engineer at IS3L for the Greenauto project, investigating neural network applications to industrial HRI. He is currently a Ph.D. candidate for a degree in electrical and intelligent systems engineering. His contributions are in industrial HRI, and he has outcomes in progress in affective computing. His research interests include artificial intelligence, HRI and its acceptance, affective computing, and theory of mind.

Ana Simões is a Senior Researcher and consultant in the Centre for Enterprise of Systems Engineering from INESC TEC in several R&D projects, funded by different types of entities, in the areas of technology adoption, innovation process, design, and development of advanced training programs, and healthcare planning. Responsible for the training area in the Industry and Innovation Lab (iiLab) at INESC TEC. Author of several publications in international and indexed journals and conferences in the field of management, technology management, and healthcare planning/management.

Paulo Menezes is affiliated with the Department of Electrical and Computer Engineering at the University of Coimbra, where he currently serves as a member of the scientific committee and board. As a lecturer, he is responsible for courses on Computer Architecture, Computer Graphics and Augmented Reality, and Interactive Systems and Robots. Additionally, he is a senior researcher at the Institute of Systems and Robotics, leading the Immersive Systems and Sensory Stimulation Laboratory and coordinating activities within the Mobile Robotics Laboratory. His research focuses on Affective Computing, Human-Robot Interaction, Virtual and Augmented Reality, and Human Behaviour Analysis.